Integration

Definition:

$$\int f'(x)\,dx = f(x) + c$$

Properties:

i. $\int n\,f(x)\,dx = n\int f(x)\,dx$, where n is a constant

ii. $\int f_1(x) \pm f_2(x)\,dx = \int f_1(x)\,dx \pm \int f_2(x)\,dx$

Useful trigonometric formulas:

- $\sin^2 x + \cos^2 x = 1$

  $1 + \tan^2 x = \sec^2 x$

  $1 + \cot^2 x = \operatorname{cosec}^2 x$

- $\sin 2x = 2\sin x \cos x$

  $\cos 2x = \cos^2 x - \sin^2 x$

  $= 2\cos^2 x - 1 \quad \rightarrow \cos^2 x = \frac{1}{2}(1 + \cos 2x)$

  $= 1 - 2\sin^2 x \quad \rightarrow \sin^2 x = \frac{1}{2}(1 - \cos 2x)$

- $\cos A \cos B = \frac{1}{2}\left[\cos(A - B) + \cos(A + B)\right]$

  $\sin A \sin B = \frac{1}{2}\left[\cos(A - B) - \cos(A + B)\right]$

  $\sin A \cos B = \frac{1}{2}\left[\sin(A - B) + \sin(A + B)\right]$

Scanned with CamScanner

Rules:

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + c, \qquad n \neq -1$$

$$\int u' (u)^n \, dx = \frac{(u)^{n+1}}{n+1} + c, \qquad n \neq -1$$

1▶

$$\int \frac{u'}{u} \, dx = \ln|u| + c$$

$$\int u' e^u \, dx = e^u + c$$

$$\int u' a^u \, dx = \frac{a^u}{\ln a} + c$$

$$\int \frac{u'}{\sqrt{u}} \, dx = 2\sqrt{u} + c$$

2▶

$$\int u' \sin u \, dx = -\cos u + c$$

$$\int u' \cos u \, dx = \sin u + c$$

$$\int u' \tan u \, dx = \ln|\sec u| + c$$

$$\int u' \cot u \, dx = \ln|\sin u| + c$$

$$\int u' \sec u \, dx = \ln|\sec u + \tan u| + c$$

$$\int u' \operatorname{cosec} u \, dx = -\ln|\operatorname{cosec} u + \cot u| + c$$

$$\int u' \sec u \tan u \, dx = \sec u + c$$

$$\int u' \operatorname{cosec} u \cot u \, dx = -\operatorname{cosec} u + c$$

Scanned with CamScanner

Sheet (1) Part (I)

Lec. Problems:

(*) $I = \int 3\sqrt{x}\, dx$

$= 3\int x^{1/2}\, dx = 3\,\frac{x^{3/2}}{3/2} + C$

(*) $I = \int \frac{1}{x+5}\, dx$

$= \ln|x+5| + C$ $\left(\int \frac{u'}{u}\, dx = \ln|u| + C\right)$

(*) $I = \int 5\, dx$

$= 5x + C$

(1) $I = \int x + x^2 + 3\, dx$

$= \frac{x^2}{2} + \frac{x^3}{3} + 3x + C$

Scanned with CamScanner

(2) $I = \int \frac{1}{\sqrt[4]{x}} \, dx$

$= \int \frac{1}{x^{1/4}} \, dx = \int x^{-1/4} \, dx = \frac{x^{3/4}}{3/4} + C$

(3) $I = \int (x^2 + 2)^2 \, dx$

$= \int (x^4 + 4x^2 + 4) \, dx$

$= \frac{x^5}{5} + 4\frac{x^3}{3} + 4x + C$

(4) $I = \int \left(\frac{1}{x^2} + \frac{4}{x\sqrt{x}} - 2 \right) dx$

$= \int (x^{-2} + 4x^{-3/2} - 2) \, dx$

$= \frac{x^{-1}}{-1} + 4\frac{x^{-1/2}}{-1/2} - 2x + C$

Scanned with CamScanner

(7) $I = \int (e^{2x})^2 \, dx$

$= \int e^{4x} \, dx$

$= \frac{1}{4} \int 4 \, e^{4x} \, dx$ $\quad \left(\frac{1}{4} \times 4 = 1\right)$

$= \frac{e^{4x}}{4} + C$ $\quad \left(\int u' e^u \, dx = e^u + c\right)$

(9) $I = \int \frac{1}{x} \cos(\ln x) \, dx$

$= \sin(\ln x) + C$ $\quad \left(\int u' \cos u \, dx = \sin u + c\right)$

(10) $I = \int \frac{\sec \sqrt{x}}{\sqrt{x}} \, dx$

$= 2 \int \frac{1}{2\sqrt{x}} \sec \sqrt{x} \, dx$ $\quad \left(2 \times \frac{1}{2} = 1\right)$

$= 2 \ln \left| \sec \sqrt{x} + \tan \sqrt{x} \right| + C$

Scanned with CamScanner

(5) $I = \int \frac{x}{\tan x^2} \, dx$

$= \int x \cot x^2 \, dx$

$= \frac{1}{2} \int 2x \cot x^2 \, dx$

$\int u' \cot u \, dx = \ln|\sin u| + C$

$= \frac{1}{2} \ln|\sin x^2| + C$

(8) $I = \int (e^{4x} + 5^x) \, dx$

$= \int e^{4x} \, dx + \int 5^x \, dx$

$= \frac{1}{4} \int 4 e^{4x} \, dx + \int 5^x \, dx$

$= \frac{e^{4x}}{4} + \frac{5^x}{\ln 5} + C$

$\int u' a^u \, dx = \frac{a^u}{\ln a} + C$

(*) $I = \int 2x \, (x^2 - 5)^7 \, dx$

$= \frac{(x^2-5)^8}{8} + C$

$\int u' (u)^n \, dx = \frac{(u)^{n+1}}{n+1} + C$

Scanned with CamScanner

(11) $I = \int x(7x^2-4)^6 \, dx$

$= \frac{1}{14}\int 14x \, (7x^2-4)^6 \, dx$

$= \frac{1}{14} \frac{(7x^2-4)^7}{7} + C$ $\quad \int u'(u)^n dx = \frac{(u)^{n+1}}{n+1} + C$

(*) $I = \int \frac{4x^3+6}{x^4+6x-1} \, dx$

$= \ln|x^4+6x-1| + C$ $\quad \int \frac{u'}{u} dx = \ln|u| + C$

(13) $I = \int \frac{1-2x^3}{\sqrt{2x-x^4+1}} \, dx$

$= \frac{1}{2}\int \frac{2-4x^3}{\sqrt{2x-x^4+1}} \, dx$ $\quad \int \frac{u'}{\sqrt{u}} dx = 2\sqrt{u} + C$

$= \frac{1}{2} \times 2\sqrt{2x-x^4+1} + C$

Scanned with CamScanner

(18) $I = \int \frac{1}{x(\ln x+1)^3}\,dx$

$= \int \frac{1}{x(\ln x+1)^3}\,dx$

$= \int \frac{1}{x}(1+\ln x)^{-3}\,dx$

$\int u'(u)^n\,dx = \frac{(u)^{n+1}}{n+1} + c$

$= \frac{(\ln x+1)^{-2}}{-2} + C$

---

(19) $I = \int \frac{(1+\cot x)^3}{1-\cos^2 x}\,dx$

$= \int \frac{(1+\cot x)^3}{\sin^2 x}\,dx$

$\cos^2 x + \sin^2 x = 1$
$\therefore 1-\cos^2 x = \sin^2 x$

$= \int \operatorname{cosec}^2 x\,(1+\cot x)^3\,dx$

$= -\int -\operatorname{cosec}^2 x\,(1+\cot x)^3\,dx$

$= -\frac{(1+\cot x)^4}{4} + C$

Scanned with CamScanner

(12) $I = \int \frac{x^2}{20x^3+4}\,dx$

$= \frac{1}{60}\int \frac{60x^2}{20x^3+4}\,dx$

$\int \frac{u'}{u}\,dx = \ln|u| + C$

$= \frac{1}{60}\ln|20x^3+4| + C$

---

(15) $I = \int \sin 2x \cos^6 2x \,dx$

$= \int \sin 2x \,(\cos 2x)^6\,dx$

$= -\frac{1}{2}\int -2\sin 2x\,(\cos 2x)^6\,dx$

$= -\frac{1}{2}\,\frac{(\cos 2x)^7}{7} + C$

Scanned with CamScanner

(17) $I = \int \tan x \cos^5 x \, dx$

$= \int \tan x \, (\cos x)^5 \, dx$

$= \int \frac{\sin x}{\cancel{\cos x}} (\cos x)^5 \, dx$

$= \int \sin x \, (\cos x)^4 \, dx$ $\left\{ \int u' (u)^n dx = \frac{(u)^{n+1}}{n+1} + C \right.$

$= - \int - \sin x \, (\cos x)^4 \, dx$

$= - \frac{(\cos x)^5}{5} + C$

(16) $I = \int \sec^8 x \tan x \, dx$

$= \int \tan x \, (\sec x)^8 \, dx$

$= \int \sec x \tan x \, (\sec x)^7 \, dx$

$= \frac{(\sec x)^8}{8} + C$

Scanned with CamScanner